- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

14 A man runs at a constant speed of $4\,\text{m s}^{-1}$ along a straight horizontal road. A woman is standing on a bridge that spans the road. At the instant that the man passes directly below the woman she throws a ball with initial speed $u\,\text{m s}^{-1}$ at $\alpha°$ above the horizontal. The path of the ball is directly above the road. The man catches the ball 2.4 s after it is thrown. At the instant the man catches it, the ball is 3.6 m below the level of the point of projection.

(a) Explain what it means that the ball is modelled as a particle. **[1]**

(b) Find the vertical component of the ball's initial velocity. **[2]**

(c) Find each of the following.

- The value of u
- The value of α

[4]